Pima Indians Diabetes Prediction (Logistic Regression)

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Default Baseline Metrics (Threshold = 0.50)	
Accuracy:	**0.734 (73.4%)**
Recall (Sensitivity):	**0.704 (38 of 54 detected)**
Precision (PPV):	**0.603**
ROC-AUC Score:	**0.813**
False Negatives (Missed):	**16 patients**

Clinically Tuned Metrics (Threshold = 0.38)	
Accuracy:	**0.714**
Recall (Sensitivity):	**0.870 (47 of 54 detected)**
Precision (PPV):	**0.573**
ROC-AUC Score:	**0.813**
False Negatives (Missed):	**7 patients (cut by 56%)**

VISUAL DIAGNOSTICS: CONFUSION MATRICES AND THRESHOLD TRADE-OFF

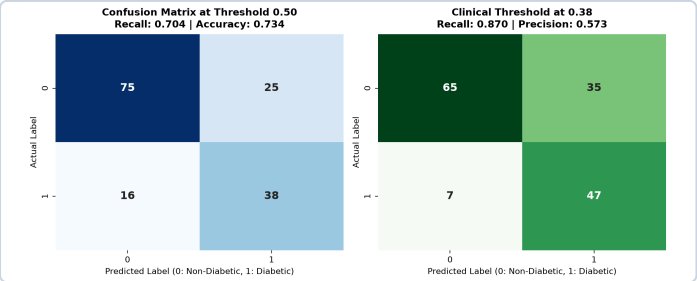


Figure 1: Confusion Matrices at Threshold 0.50 vs Tuned 0.38

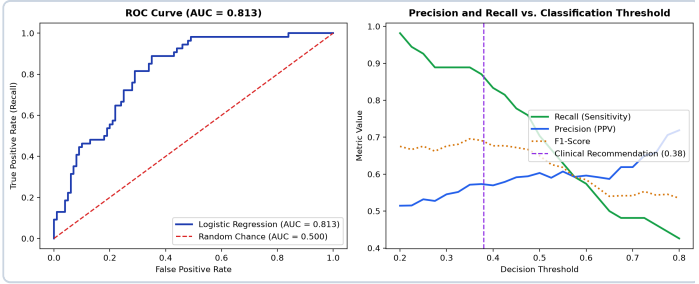


Figure 2: ROC Curve (AUC 0.813) and Precision-Recall vs Threshold

ANTI-LEAKAGE DATA ENGINEERING DECISIONS

- **Handling Biologically Implausible Zeroes:** In medical science, blood glucose, diastolic blood pressure, triceps skinfold, insulin, and BMI cannot equal zero. These indicate missing measurements. I replaced zeroes in these five attributes with NaN.
- **Strict Anti-Leakage Separation:** To prevent test set contamination, median imputation and standard scaling were fitted strictly on the 80% training split (614 patients) and transformed onto the 20% test split (154 patients).
- **Class Balancing:** Because non diabetic patients outnumber diabetic patients roughly 2 to 1 (65% to 35%), I trained Logistic Regression with class_weight='balanced' to ensure the loss function penalizes errors on diabetic individuals appropriately.

CLINICAL DISCUSSION: WHY LOWERING THE THRESHOLD SAVES LIVES

The Medical Trade-Off in Plain English:
In disease screening, a False Negative is far more harmful than a False Positive. A False Negative means a diabetic individual is told they are healthy, delaying essential medical treatment and risking diabetic ketoacidosis, organ damage, or cardiovascular events. Conversely, a False Positive simply means an otherwise healthy individual undergoes an inexpensive confirmatory laboratory blood test. By shifting the classification threshold from 0.50 to 0.38, our model recall climbs from 70.4% to 87.0%, detecting 9 additional diabetic patients and reducing missed cases from 16 down to 7.